
Week 4, Lecture 2 Handout: Mobile-First Responsive Design & Media Queries

Why Mobile-First Design? (The Professional Rule)

Historically, developers built complex desktop layouts first, then tried to squish and patch them onto mobile devices. This caused massive layout bugs and heavy site rendering times. [1, 2]

Modern global tech entities use a **Mobile-First Workflow**. You write the default CSS rules for the smallest smartphone screens first—using clean, vertical, single-column blocks. Then, you use **Media Queries** to layer on layout changes as the screen viewport expands onto tablets and large desktop monitors. [3, 4, 5, 6, 7]

The Core Responsive Toolbox

1. The Critical Viewport Meta Tag

Every responsive website requires this single tag embedded inside the HTML `<head>`. Without it, mobile browsers simulate a tiny desktop screen, shrinking your text until it is unreadable: [8, 9, 10, 11]

```
<meta name="viewport" content="width=device-width, initial-scale=1.0">
```

- `width=device-width`: Tells the page to match the screen's literal width in pixels.
- `initial-scale=1.0`: Sets the initial zoom level to 100%. [12, 13, 14, 15]

2. The Media Query Conditional Engine

A Media Query is a specialized conditional gate. The browser reads it as: *"If the screen width matches these specific pixel conditions, activate these internal styling rules. Otherwise, completely ignore them."* [16]

```
/* DEFAULT MOBILE RULES GO HERE (Applies to all screens 0px and wider)
*/
.card-grid { display: block; }
/* THE TABLET / DESKTOP BREAKPOINT GATEWAY */@media (min-width: 768px)
{
    /* RULES HERE ONLY ACTIVATE WHEN THE VIEWPORT IS 768px OR WIDER
    */
    .card-grid { display: grid; }
}
```